Subject-Independent Hand-Gesture Recognition from Surface EMG on Chained INT8 LDA Boards:

From Gesture Selection to a Bit-Exact Multi-Board Design

Vinay Hariharan Sudhindra Sahoo Yashvardhan Khandegar Digvijay Kumbhar

Department of Electrical Engineering, Indian Institute of Technology Bombay

Advisor: Prof. Laxmeesha Somappa (BioNICS group)

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Abstract

We design and verify a low-power digital classifier that recognises four hand and wrist gestures from forearm surface electromyography (sEMG) on a person it has never seen. Every design decision is settled by a subject-independent experiment on the public FORS-EMG dataset (19 subjects, 8 electrodes, 985 Hz), using leave-one-subject-out evaluation and paired per-subject statistics. Four findings shape the design. (i) The gesture vocabulary matters more than the model: choosing whole-hand and wrist gestures instead of finger postures gains 14.1 pp, and keeping all twelve gestures by merging them into four classes loses 16.9 pp on every subject. (ii) Model capacity does not: an MLP, logistic regression, a linear SVM and linear discriminant analysis (LDA) are statistically tied, and a convolutional network on the raw signal gains at most 2.7 pp for about 3,700 times the multiply-accumulates. (iii) The whole front end can be integer-only: five time-domain features built from add, compare and shift, a shift normaliser and INT8 weights cost no measurable accuracy, and a bit-exact model of the chip reaches $77.6 \pm 4.3\%$ on new users. One calibration recording per gesture raises the LDA to $86.4 \pm 2.6\%$. (iv) The LDA splits exactly across boards: each board handles up to eight electrodes and sends only three 32-bit partial sums per decision, so 8 to 24 electrodes are served by identical boards on one 256 kHz clock with one multiplier each. The Verilog reproduces the golden model on every decision of a real recording for six board configurations, and a mutation test shows that the testbench catches each of eight planted bugs.

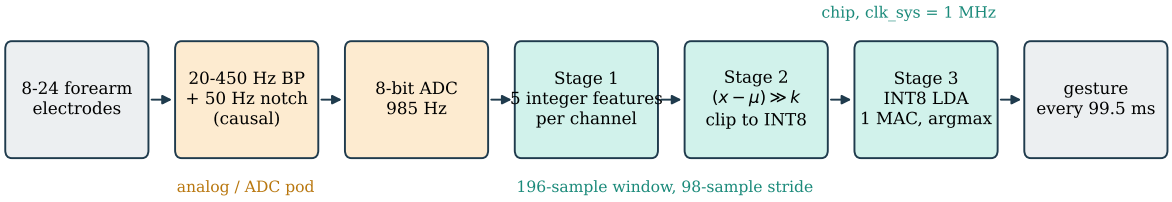


Figure 1: The classifier. The electrode pods filter and digitise the signal; the chip computes five integer features per channel, normalises them with a shift, and an INT8 LDA decides one of four gestures every 98 samples (about 100 ms).

1 Introduction

Myoelectric control turns the electrical activity of forearm muscles into commands for a prosthetic hand, an orthosis or a gesture interface. The signal is measured with surface electrodes on the skin,

so it is cheap and non-invasive; the price is that it is weak, noisy and different for every person. A practical controller has to satisfy three requirements at once.

1. **It must work for a new user.** Skin and fat thickness, electrode placement and muscle physiology change the signal from person to person. A classifier that is accurate only on the people it was trained on is of little use. We therefore report only *subject-independent* accuracy.
2. **It must run continuously on a battery.** This favours integer arithmetic, few multiplications, small memories and a slow clock.
3. **It must feel immediate.** The optimal delay between the user's intent and the device's response is about 100–125 ms [7]; we decide every 100 ms.

Pattern-recognition control with time-domain features and LDA has been the reference approach for three decades [2, 3, 4]. The questions this work answers are which parts of that recipe survive the constraints above, what a larger model would add, and how to build the result so that it scales from 8 to 24 electrodes without redesign.

Contributions.

- A complete, subject-independent design study on FORS-EMG [1] in which every choice (gestures, classifier, arithmetic, features, normaliser, calibration) is backed by a paired test (Sections 4–8).
- A direct comparison of *dropping* gestures against *merging* all twelve into fewer classes, with the grouping chosen inside each fold from the training subjects only (Section 4.2).
- Evidence that a raw-signal CNN gains at most a few points over the LDA, before and after per-user calibration, at three orders of magnitude more arithmetic (Sections 7 and 8).
- A fully integer front end (add, compare, shift and a leading-one detector) that is non-inferior to floating-point features (Section 6.2).
- An exact decomposition of the LDA over a chain of identical boards that exchanges only $K - 1$ partial sums, with a self-configuring chain protocol, one 256 kHz clock and one multiplier per board (Section 9).
- Verification by a bit-exact golden model, a self-checking unit test, a mutation test and a whole-chip test on real data for six board splits (Section 10).

2 Background

Features and LDA. Hudgins *et al.* [2] introduced a small set of time-domain (TD) features (mean absolute value, zero crossings, slope sign changes, waveform length) computed over short windows, and Englehart and Hudgins [3] showed that these features with an LDA give a robust, real-time controller with very little storage. Phinyomark *et al.* [5] later showed that many published TD features are redundant, which motivates our feature culling.

Generalisation and robustness. Clinical use is limited less by accuracy on a fixed session than by changes between users, sessions and arm positions [4]. The limb-position effect is well documented [6]; FORS-EMG [1] was recorded in three forearm orientations precisely to study it.

Integer inference. Symmetric INT8 weights with INT32 accumulators are the standard way to run linear layers and neural networks without floating point [8]. Mitchell's logarithm approximation [9] gives cheap squares and products in hardware.

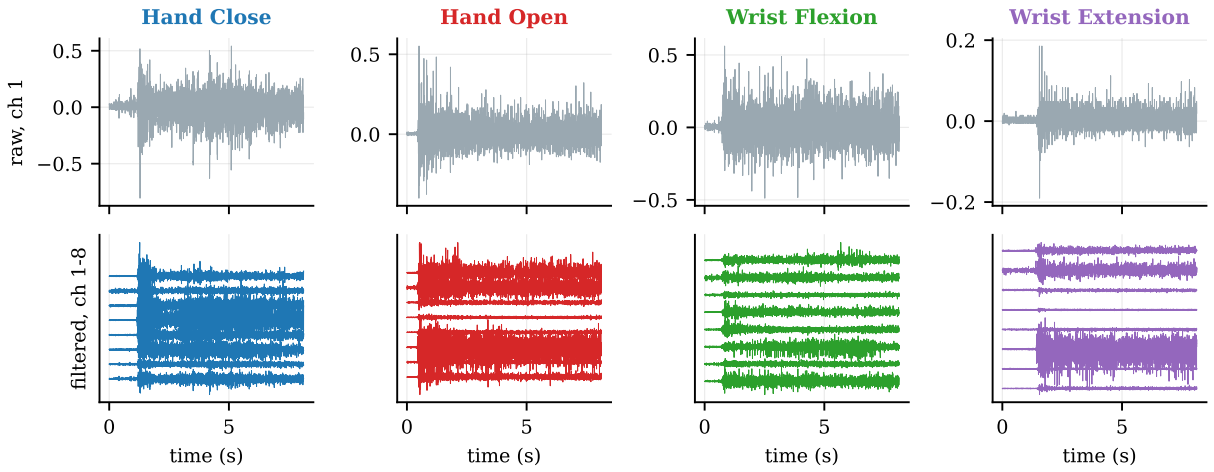


Figure 2: One recording of each chosen gesture (subject 2, rest orientation). Top: raw channel 1. Bottom: all eight channels after the causal filter. The gestures differ mainly in *which* channels are active, which is what per-channel amplitude features capture.

3 Dataset and evaluation protocol

Data. FORS-EMG [1] contains 19 subjects, 8 forearm electrodes sampled at 985 Hz, 12 gestures, 3 forearm orientations (rest, supination, pronation) and 5 trials per gesture and orientation. Each recording has 8000 samples (8.1 s). Every gesture class pools all three orientations. Figure 2 shows one recording of each of the four gestures finally used.

Signal chain. A fourth-order Butterworth band-pass (20–450 Hz) and a 50 Hz notch ($Q = 30$), run *causally* as cascaded biquads, because hardware cannot use future samples. Early experiments used zero-phase filtering; every result that bears on the chip uses the causal filter. The filter removes baseline drift and the mains line and keeps the EMG band (Figure 3).

Windows. 200 ms windows (196–197 samples) with a 100 ms stride, so consecutive windows share half their samples. The chip uses exactly 196 samples (two banks of 98, Section 9.2).

Split. Leave-one-subject-out (LOSO): in each fold one subject is held out and never used for training, normalisation, quantisation scales or any other choice. Within the training subjects, trials 1–4 train and trial 5 validates. Windows are cut *after* the split. An early version cut windows first; because neighbouring windows overlap, validation windows leaked into training and produced a false accuracy ceiling of about 63% that did not depend on model capacity.

Statistics. Accuracy is the mean over held-out subjects with a 95% confidence interval (Student t). Two methods are compared subject by subject: the mean paired difference, its 95% interval, the Wilcoxon signed-rank p-value and the number of subjects up and down. A difference is called real only if its interval excludes zero; non-inferiority uses a margin of -1 pp.

Subject count. The copy of the dataset in the project repository lacks subject 1. Runs made from it use 18 subjects; runs on the full dataset use 19. Each table states which.

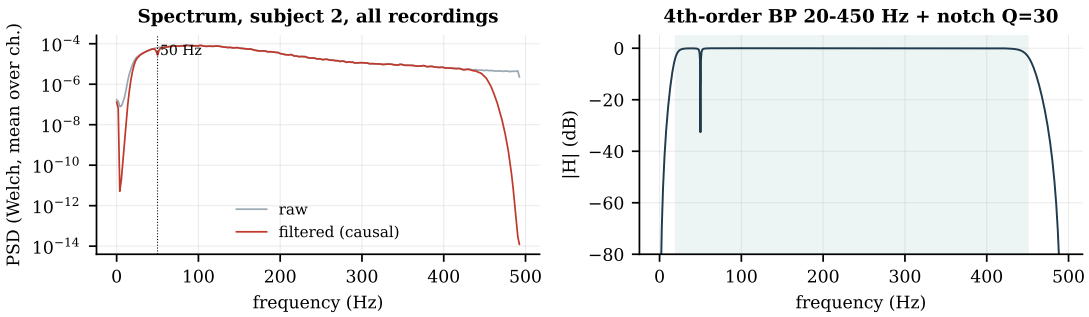


Figure 3: Left: power spectrum before and after the causal filter. Right: the filter's magnitude response.

Table 1: Accuracy by gesture set (LOSO, 19 subjects, 64 floating-point features).

Set	Model	FP32 %	INT8 %	INT8 loss (pp)	MACs	Bytes
A (4, hand + finger)	MLP 64-4	65.17	65.28	−0.12	256	272
B (4, hand + wrist)	MLP 64-16-4	79.69	79.38	0.32	1088	1168
D (6)	MLP 64-16-6	69.23	69.21	0.02	1120	1208
	Logistic reg.	68.95	68.74	0.21	384	408
C (12)	MLP 64-16-12	48.99	48.37	0.63	1216	1328
	Logistic reg.	49.71	49.28	0.43	768	816

4 Which gestures, and how many

4.1 Gesture kinds

The twelve FORS-EMG gestures are of three kinds, driven by different muscles: whole-hand movements (Hand Close, Hand Open), wrist movements (flexion, extension, radial and ulnar deviation), and finger postures (Index, Index-Little, Peace, Thumb Up, Thumb-Little, Right Angle). Whole-hand and wrist movements use large, superficial muscles; finger postures use the deep, overlapping compartments of the finger flexors, which eight surface electrodes see only weakly.

Our first set, A = {Hand Close, Hand Open, Index, Index-Little}, stopped at about 65% for every model. Its confusion matrix showed that Index ↔ Index-Little alone caused about 40% of all errors. Set B = {Hand Close, Hand Open, Wrist Flexion, Wrist Extension} replaces the two finger postures with two wrist movements and gains +14.1 pp (paired t-test, 95% interval [+11.3, +16.9], $p = 3.8 \times 10^{-9}$), more than any later model or hardware choice (Table 1, Figure 4).

How many gestures. Larger vocabularies cost accuracy. To find the best six, we ranked all $\binom{12}{6} = 924$ subsets by their worst confused pair in the 12-class confusion matrices. The best subset (set D) was predicted at 66.4% before training and measured at 69.0%; all twelve gestures give 48.8%. In set D, Right Angle alone caused 44.5% of the errors. With twelve classes, the MLP and logistic regression agreed on nine of their ten most confused pairs, so the errors come from the signals, not from the model.

4.2 Merging all twelve gestures instead of dropping eight

Dropping eight gestures leaves two thirds of the recordings unused. The alternative is to keep every gesture and merge them into four classes. To compare the two fairly we used the chip pipeline of Section 6.2 (8-bit ADC, streamed 196-sample windows, integer features, shift normaliser, INT8 LDA) for both.

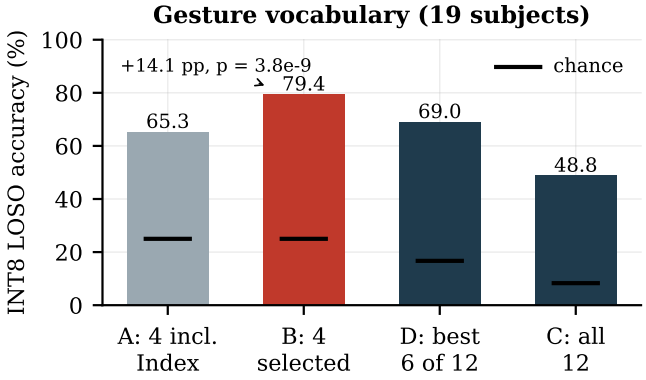


Figure 4: INT8 accuracy by gesture set (19 subjects); black bars mark chance.

Table 2: Dropping versus merging (LOSO, 19 subjects, chip INT8 LDA). The paired difference is against dropping.

Approach	Gestures used	Class rows	Accuracy %	vs drop (pp, 95% CI, up/down)
Drop to HC, HO, WF, WE	4 of 12	4	77.82 ± 4.10	—
Merge, 4 groups of 3, 4-class LDA	all 12	4	60.91 ± 4.55	-16.9 $[-20.3, -13.6]$, 0/19
Merge, 4 groups of 3, 12-class mapped	all 12	12	63.07 ± 5.07	-14.8 $[-18.2, -11.3]$, 0/19
Earlier 6-group map, 6-class LDA	all 12	6	72.38 ± 4.11	-5.5 $[-8.0, -2.9]$, 1/18
Earlier 6-group map, 12-class mapped	all 12	12	70.71 ± 4.62	-7.1 $[-10.4, -3.9]$, 2/17
12 gestures, no merging	all 12	12	46.41 ± 5.55	—

Choosing the grouping without the test subject. Inside each outer fold, a 12-class LDA is cross-validated over the training subjects only (three folds of subjects), giving a 12×12 confusion matrix N. Among all $12!/(3!^4\,4!) = 15{,}400$ ways to split the gestures into four groups of three, we choose the split $\mathcal{G}$ that maximises the share of windows predicted inside their own group,

$$\mathcal{G}^\star = \arg\max_{\mathcal{G}} \frac{1}{\sum_{ij} N_{ij}} \sum_{g \in \mathcal{G}} \sum_{i \in g} \sum_{j \in g} N_{ij}. \tag{1}$$

Groups of equal size keep the classes balanced and stop the search from lumping every hard gesture into one large class. The merged model is evaluated in two ways: a four-class LDA trained on the merged labels (same chip), and the 12-class LDA whose decision is mapped to its group (three times the class rows). We also evaluate the fixed six-group map used in an earlier stage of the project (G1 = Hand Close, Index, Index-Little, Peace, Right Angle, Thumb Up; G2 = Hand Open, Wrist Extension; G3–G6 = Radial Deviation, Thumb-Little, Ulnar Deviation, Wrist Flexion).

Results. Dropping wins on every subject (Table 2, Figure 5). The chosen groupings are physiologically sensible: Hand Open and Wrist Extension fall into the same group in 19 of 19 folds, and Hand Close joins both thumb gestures in 17 of 19, while the groups of finger postures change from fold to fold. Two effects explain the loss. First, the finger postures stay confusable however they are grouped (twelve-class accuracy 46.4%). Second, a merged class is a mixture of different muscle patterns, which an LDA, with one Gaussian per class, fits poorly: mapping a 12-class decision to the groups beats training on merged labels by $+2.2$ pp ($[+0.5, +3.8]$, $p = 0.02$). The earlier six-group map reaches 72.4%, but its largest group holds half of all windows, so answering G1 every time already scores 50%. **From here on the classifier uses the four gestures of set B.**

A caveat on selection. The ranking that chose set D, and the confusion data that confirmed set B, came from LOSO confusion matrices that include the test subjects, which is mildly optimistic.

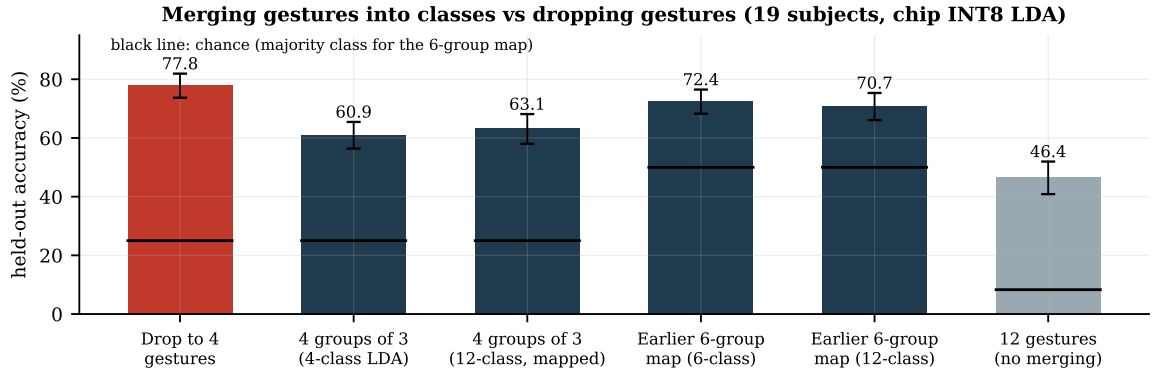


Figure 5: Merging versus dropping. Bars: mean over 19 held-out subjects with 95% intervals; black lines: chance (for the six-group map, the share of the largest group, 50%).

Table 3: Eight classifiers on set B (LOSO, 19 subjects). "vs MLP" is a paired test.

Model	FP32 %	INT8 %	INT8 loss	MACs	Bytes	vs MLP
MLP 64-16-4	79.65	79.57	0.08	1088	1168	—
Gradient boosting	79.63	79.26	0.37	0	19,200	tied ($p = 0.78$), over budget
Logistic regression	79.33	79.19	0.14	256	272	tied ($p = 0.44$)
Linear SVM	79.31	78.30	1.01	256	272	tied ($p = 0.29$)
LDA	77.91	77.87	0.04	256	272	tied ($p = 0.09$)
Random forest	77.84	77.43	0.41	0	53,145	worse ($p = 0.046$)
Extra trees	76.29	76.04	0.25	0	38,015	worse ($p = 0.0009$)
Decision tree	73.69	73.17	0.52	0	5,055	worse ($p = 0.0002$)

The merge experiment above chooses its groups inside each fold and is free of this bias. Set B was chosen first on muscle grounds, and its 14 pp gain is far larger than any plausible selection effect.

5 Classifier

5.1 Eight classifiers on the same features

All models in Table 3 see the same 64 floating-point features (eight TD types on eight channels), the same folds and the same INT8 scheme, under a 10 KB memory budget. The MLP has a ReLU hidden layer, dropout 0.2, Adam with a cosine learning rate and early stopping on trial 5; the LDA uses a shared covariance with Ledoit–Wolf shrinkage [10]; trees have depth 10, forests 25 trees of depth 8, gradient boosting 60 rounds of depth 4.

The four linear-capacity models and the MLP are statistically tied; trees are worse or exceed the budget (Figure 6). On set A, five MLP widths from a single linear layer to 64-128-4 differ by 1.3 pp, less than the ±5.9 pp interval of any one of them, with 34 times the memory at the top (Figure 7). The same tie between the MLP and logistic regression appears at 4, 6 and 12 gestures. **Model capacity is not the bottleneck; the information in eight surface channels is.**

5.2 The LDA, and why we chose it

With class means μ_k, one shared covariance Σ and priors π_k, the LDA assigns a feature vector q to the class with the largest linear score

$$s_k(q) = w_k^\top q + b_k, \qquad w_k = \Sigma^{-1}\mu_k, \qquad b_k = -\tfrac{1}{2}\,\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k. \tag{2}$$

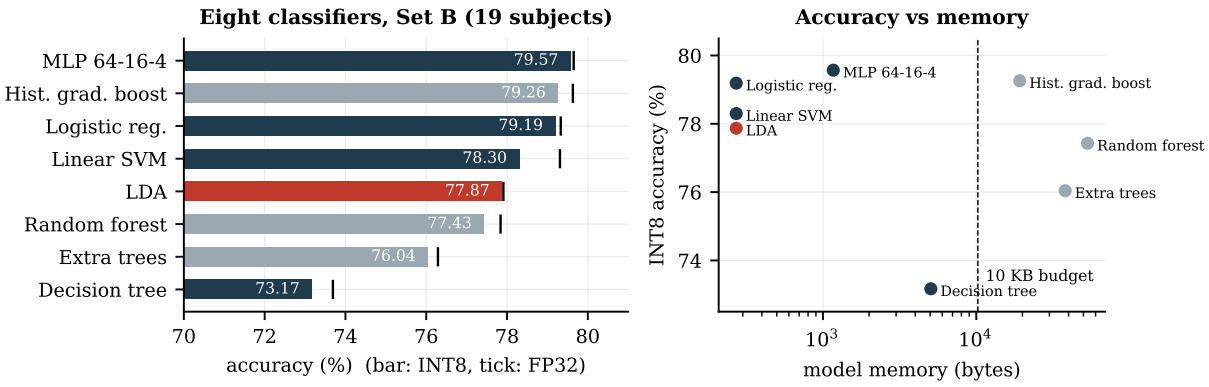


Figure 6: Left: INT8 accuracy (bars) and FP32 accuracy (ticks) of the eight classifiers. Right: accuracy against memory; grey models break the 10 KB budget, red is the LDA used on the chip.

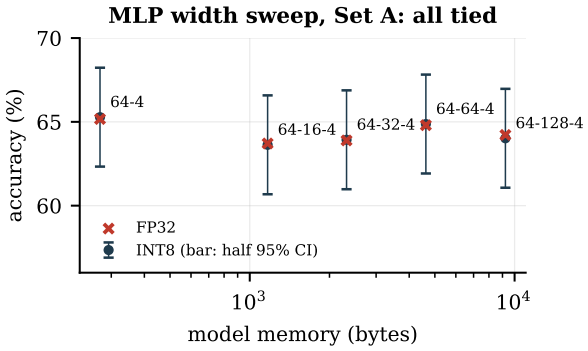


Figure 7: MLP width against memory on set A: 34 times the memory buys no accuracy.

Logistic regression had a slightly better point estimate, but we chose the LDA because: (1) its training is closed form (means and one covariance), so retraining for a new user or a new electrode layout is a few matrix operations and no optimiser, which Section 8 exploits; (2) its weights are stored in writable memory, so a new model needs no new silicon; (3) its scores are log-likelihoods up to a shared constant. It costs 1.7 pp against the MLP, inside the tie, for 4.3 times fewer multiply-accumulates (MACs) and bytes.

5.3 INT8 arithmetic

Weights and activations are symmetric per-tensor INT8 with the scale set at the 99.9th percentile of the magnitudes; biases are INT32 on the product scale $s_W s_x$, so they add directly into the accumulator. All scales come from the training subjects. Over 28 FP32-versus-INT8 comparisons (every model and gesture set) the loss is 0.18 pp on average (median 0.11, worst 1.01 for the linear SVM, whose hinge loss produces larger weights), and 25 of 28 losses are below 0.5 pp (Figure 8). A requantisation bug (a shift of $\lceil -\log_2 M \rceil$, undefined for multipliers $M > 1$) would have cost more than 50 pp; a synthetic unit test caught it before any result was reported.

The chip needs no requantisation at all. Only the argmax is used, and if every class score shares one positive scale s, then $\arg\max_k s\, S_k = \arg\max_k S_k$. The chip therefore keeps one INT32 accumulator per class and one weight scale for all classes.

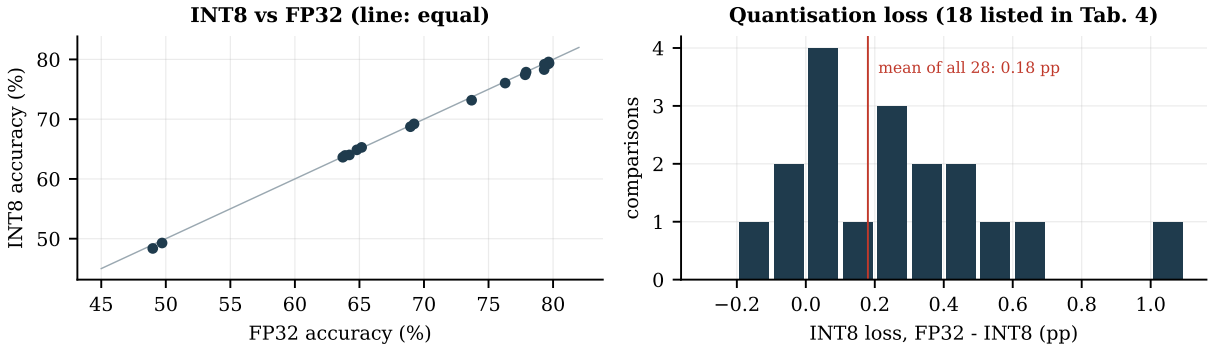


Figure 8: INT8 against FP32 for the results of Tables 1–3 (left), and the distribution of the losses (right).

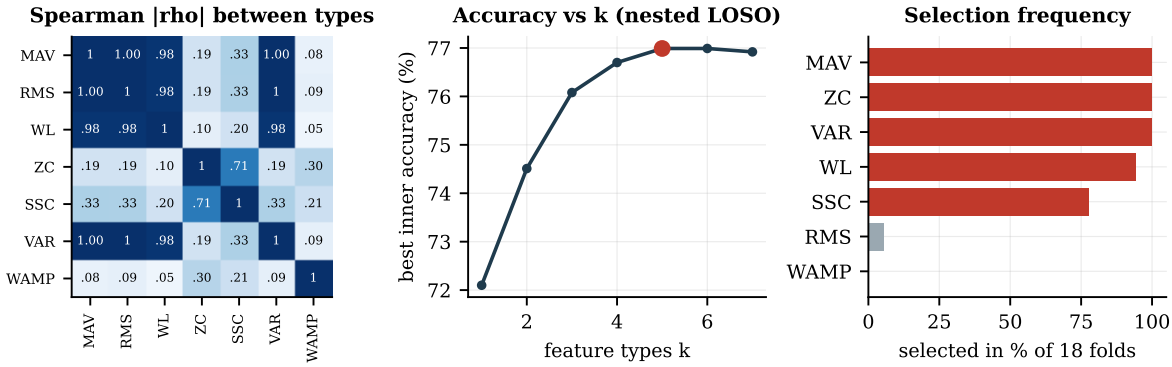


Figure 9: Feature-type selection. Left: rank correlation between types. Middle: best inner-LOSO accuracy for each number of types. Right: how often each type was selected over the 18 outer folds.

6 Features

6.1 Which feature types

sEMG is noise-like: what separates gestures is how strongly and how fast each channel's signal varies, which is exactly what TD features measure [2, 3]. We started from eight TD types per channel (64 inputs) and selected types by nested LOSO on the INT8 LDA (18 subjects): for each held-out subject, all 127 subsets of seven candidate types were scored by an inner LOSO on the other 17 subjects, and the smallest subset within one paired standard error of the best was kept. IEMG was excluded beforehand, because IEMG $= N \cdot$MAV gives an identical LDA.

MAV, RMS, WL and VAR are almost the same signal (Spearman $|\rho| \geq 0.977$, Figure 9), in line with [5]; ZC, SSC and WAMP carry different information, but WAMP saturates (98% of sample differences exceed its threshold). The selected set {MAV, WL, ZC, SSC, VAR} reaches $77.08 \pm 4.97\%$ against $77.31 \pm 5.02\%$ with all eight types ($\Delta = -0.23 \pm 0.42$ pp, $p = 0.20$, non-inferior), with 40 inputs instead of 64: 160 instead of 256 MACs and 176 instead of 272 bytes. MAV, ZC and VAR were chosen in all 18 folds, WL in 17 and SSC in 14.

6.2 An integer-only front end

The selected features need division, square roots and floating-point thresholds. We rebuilt them from add, subtract, absolute value, compare, shift and a leading-one detector. For an 8-bit sample

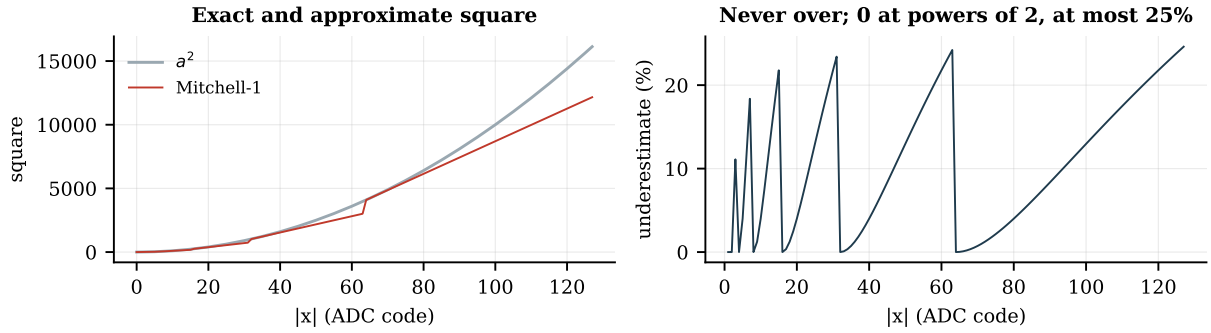


Figure 10: Mitchell's square of an 8-bit magnitude: a smooth, monotone underestimate, so the window sum stays almost perfectly correlated with the exact sum of squares.

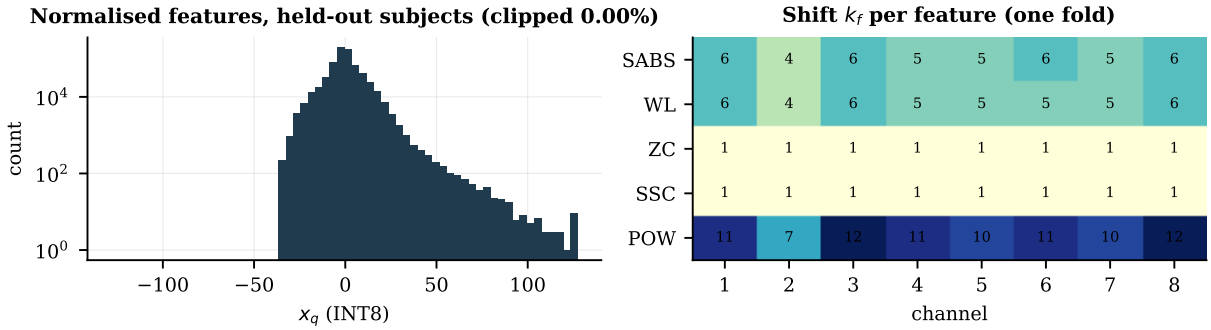


Figure 11: Left: the INT8 inputs q_f on held-out windows. Right: the shifts k_f of one fold.

x_n of one channel and a window of N samples:

$$\text{SABS} = \sum_n |x_n|, \qquad \text{WL} = \sum_n |x_n - x_{n-1}|, \tag{3}$$

$$\text{ZC} = \sum_n \big[\operatorname{sign}(x_n) \neq \operatorname{sign}(x_{n-1})\big], \qquad \text{SSC} = \sum_n \big[d_n d_{n-1} < 0\big], \quad d_n = x_n - x_{n-1}, \tag{4}$$

$$\text{POW} = \sum_n \widehat{a_n^2}, \qquad \widehat{a^2} = a \cdot 2^{p+1} - 2^{2p}, \quad p = \lfloor \log_2 a \rfloor, \; a = |x_n|. \tag{5}$$

SABS replaces MAV (the factor $1/N$ is absorbed by the normaliser), ZC is a sign-bit XOR without a dead-band, SSC counts strict changes of slope sign (only the sign of the previous difference is stored), and POW replaces VAR. The square in POW is Mitchell's one-step square [9]: one leading-one detector, two shifts and one subtraction. Writing $a = 2^p(1+f)$ with $0 \le f < 1$ gives $a^2 - \widehat{a^2} = 2^{2p} f^2 \ge 0$, so it never overestimates, is exact at powers of two and errs by at most 25% (Figure 10); the summed feature still has correlation $r = 0.9999$ with $\sum x^2$.

Shift normaliser. Z-scoring and input quantisation are replaced by

$$q_f = \operatorname{clip}\Big(\big((x_f - \mu_f) + 2^{k_f - 1}\big) \gg k_f,\; -128,\; 127\Big), \tag{6}$$

with an integer μ_f (the rounded training mean) and the smallest $k_f \ge 1$ that never clips on the training subjects. The LDA is invariant to a positive rescaling of any one feature, so replacing the standard deviation by a power of two costs nothing in principle; rounding and clipping are the only approximations. Count features need $k = 1$, amplitude features 4–6 and power features 7–12 (Figure 11).

Result. Six integer variants were compared with the floating-point reference on the same folds (Table 4, Figure 12). Every variant that keeps a power feature is non-inferior; dropping power costs

Table 4: Integer feature variants, INT8 LDA, LOSO (18 subjects). Δ: paired difference to the reference R, with the shift normaliser.

	Power feature / ZC	Standard normaliser %	Shift normaliser %	Δ vs R (pp, 95% CI)
R	floating-point reference	77.45	–	–
H1	exact $\sum x^2$	77.33	77.63	+0.19 [−0.36, +0.73]
H2	Mitchell, two steps	77.29	77.56	+0.11 [−0.41, +0.63]
H3	Mitchell, one step	77.26	77.39	−0.05 [−0.57, +0.47]
H4	$(\sum \lvert x \rvert)^2$	77.10	77.14	−0.31 [−0.88, +0.27]
H5	no power feature	75.38	75.54	−1.91 [−2.94, −0.87]
H6	Mitchell, one step; ZC without dead-band	77.29	77.39	−0.05 [−0.59, +0.48]

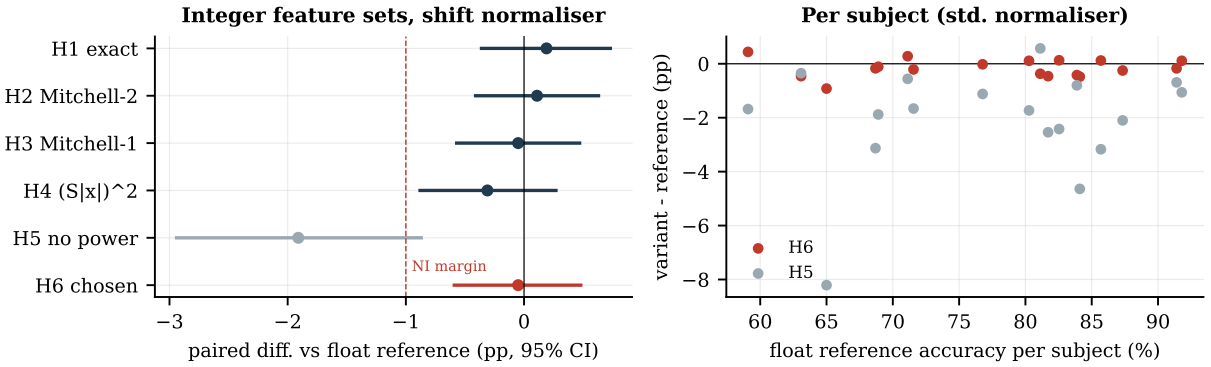


Figure 12: Left: paired difference of each integer variant to the floating-point reference; the dashed line is the non-inferiority margin. Right: per-subject differences for H6 and for H5, which drops the power feature.

1.9 pp and fails, so a power feature is needed; the shift normaliser costs nothing against z-scoring. We chose H6, the cheapest variant. On 19 subjects the integer front end gives 77.79±4.39% against 77.88% for floating-point features.

6.3 The chip's own numbers

A golden model reproduces every integer of the chip: an 8-bit ADC, recordings streamed sample by sample (so WL, ZC and SSC use the true previous sample), 196-sample windows from two banks, the shift normaliser and an INT8 LDA with one weight scale. Trained and tested by LOSO on its own bit-exact features, it reaches $77.61 \pm 4.34\%$ against $77.83 \pm 4.11\%$ for the software INT8 LDA: $\Delta = -0.21$ pp, $[-0.72, +0.30]$, $p = 0.49$, nine subjects up and nine down (Figure 13). There is no detectable cost to the chip's arithmetic. The value ranges leave wide margins: μ_f uses at most 13 of its 22 bits, $k_f \in [1, 12]$, scores stay below 4047 in magnitude against the INT32 limit, and no feature clips.

Most remaining errors are Wrist Extension predicted as Hand Open (29%) and the reverse (18%) (Figure 14): the two movements share extensor-digitorum activity.

7 Would a convolutional network do better?

If the features carry the information, a network that learns its own features from the raw signal might find what our five types miss. Two raw-signal CNNs were trained on exactly the same windows and folds as the LDA. Every convolution is a 1-D convolution with batch normalisation and ReLU.

- **Option 2 (depthwise-separable).** A depthwise temporal convolution (8 → 32, kernel 15,

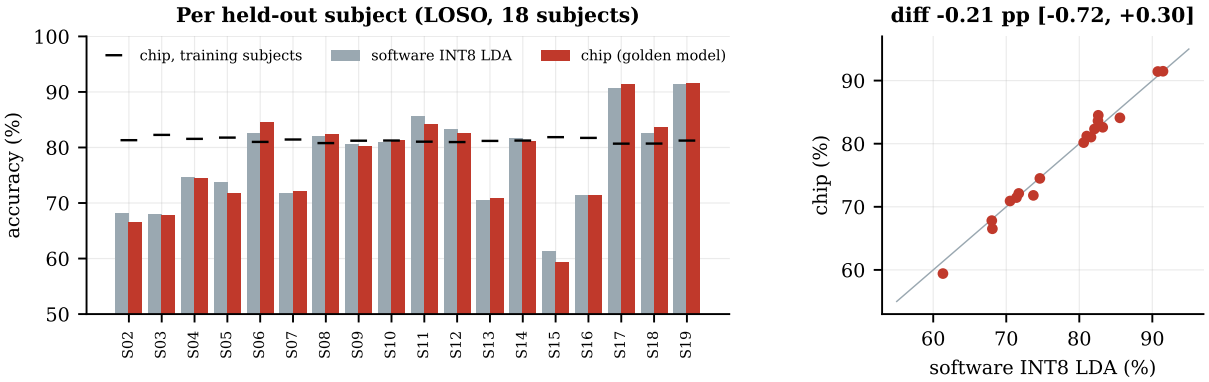


Figure 13: Held-out accuracy of the chip model and the software INT8 LDA per subject; the points lie on the diagonal.

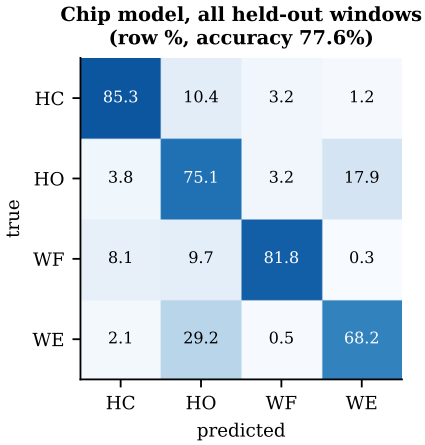


Figure 14: Confusion matrix of the chip model over all held-out windows of all folds (row %).

four filters per electrode), a pointwise mix ($32 \to 32$), three separable blocks with stride 2 ($32 \to 48 \to 64 \to 64$), global average pooling and a $64 \to 4$ output layer.

- **Option 5c (two-branch).** A temporal branch (depthwise, kernel 15) and a spatial branch (pointwise $8 \to 32$) in parallel, concatenated to 64 channels, then two separable blocks, pooling and the output layer.

Training uses AdamW, a cosine learning rate, label smoothing 0.1, dropout 0.3, at most 150 epochs and early stopping on trial 5 (patience 20). Each electrode of each training window is multiplied by a random gain e^z, $z \sim \mathcal{N}(0, 0.3^2)$, imitating differences in skin contact between people. For INT8, batch normalisation is folded into the convolution weights.

The separable CNN is not distinguishable from the LDA; the two-branch CNN is better by 2.7 pp on 15 of 19 subjects (Table 5, Figure 15). INT8 changes neither network measurably. The gain costs about 3,700 times the MACs and 74 times the parameters. A network that learns its features from raw samples finds only a little that the TD features miss, which is the same conclusion as the classifier comparison from a different direction: with eight surface electrodes, the signal, not the model, sets the limit.

Table 5: Raw-signal CNNs against the LDA (LOSO, 19 subjects, INT8). The paired difference is against the LDA.

Model	MACs / decision	Parameters	Accuracy %	vs LDA (pp, 95% CI, up/down)
LDA, 40 integer features	160	164	77.79 ± 4.39	—
CNN option 2	751,584	11,732	79.10 ± 4.10	$+1.31$ $[-1.00, +3.61]$, 14/5, $p = 0.20$
CNN option 5c	594,400	12,132	80.48 ± 3.81	$+2.68$ $[+0.76, +4.61]$, 15/4, $p = 0.008$

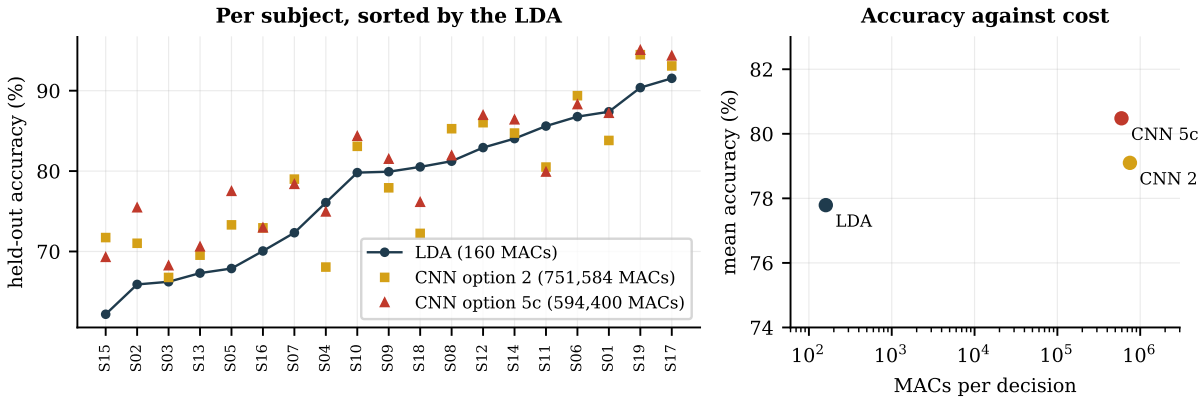


Figure 15: Left: per-subject accuracy of the LDA and the two CNNs, sorted by the LDA. Right: mean accuracy against MACs per decision (log scale).

8 Per-user calibration

The global model has never seen the user; it reaches about 78% on average and under 60% for the worst user. A short recording of the user, one repetition of each gesture (about 8 s each), can adapt it.

8.1 Calibrating the LDA

Because the LDA is closed form, calibration only blends statistics. With the global class means μ_k^g and covariance Σ^g, and the same quantities μ_k^u, Σ^u estimated from the user's recording,

$$\mu_k \leftarrow (1-\lambda)\,\mu_k^g + \lambda\,\mu_k^u, \qquad \Sigma \leftarrow (1-\lambda)\,\Sigma^g + \lambda\,\Sigma^u, \tag{7}$$

and w_k, b_k are recomputed from Equation (2): one 40×40 solve. The new weights are quantised and written to the chip over its configuration port; no hardware change is needed. With $\lambda = 0.5$, calibration on trial 1 and testing on trials 3–5 of each held-out subject (never used for calibration), the INT8 LDA rises from $78.03 \pm 4.76\%$ to $86.35 \pm 2.56\%$ on 18 subjects: $+8.3$ pp ($[+4.4, +12.3]$), all 18 subjects improve, Wilcoxon $p = 2 \times 10^{-4}$ (Figure 16, left). The worst users gain most (S15 $+30.1$ pp, S13 $+20.1$ pp), so the spread between users halves. A second calibration trial adds only 0.4 pp.

8.2 Calibrating a CNN, and the decision

A dedicated study (the project's CNN checkpoint) compared the two families after calibration on all 19 subjects, with three training seeds per CNN and five rounds per subject in which every trial serves once each as test, validation and calibration data. Its setting differs from the rest of this paper: all twelve gestures are classified and mapped to the six groups of Section 4.2, 150 ms windows with a 50 ms stride are used, only windows after the first 0.5 s of each recording (the gesture is held) are scored, and the LDA is calibrated by first re-centring the features on the user's

Table 6: CNN checkpoint: group accuracy (six groups) on held windows, INT8, 19 subjects.

Model	No calibration	1 trial (≈5 min)	2 trials	3 trials	Gain from 1 trial
LDA (re-centre + blend)	71.4	91.9	92.7	93.1	+20.5
CNN "small" (49k MACs)	75.4	92.2	–	–	+16.9
CNN "two" (452k MACs)	78.1	94.3	95.2	95.9	+16.2

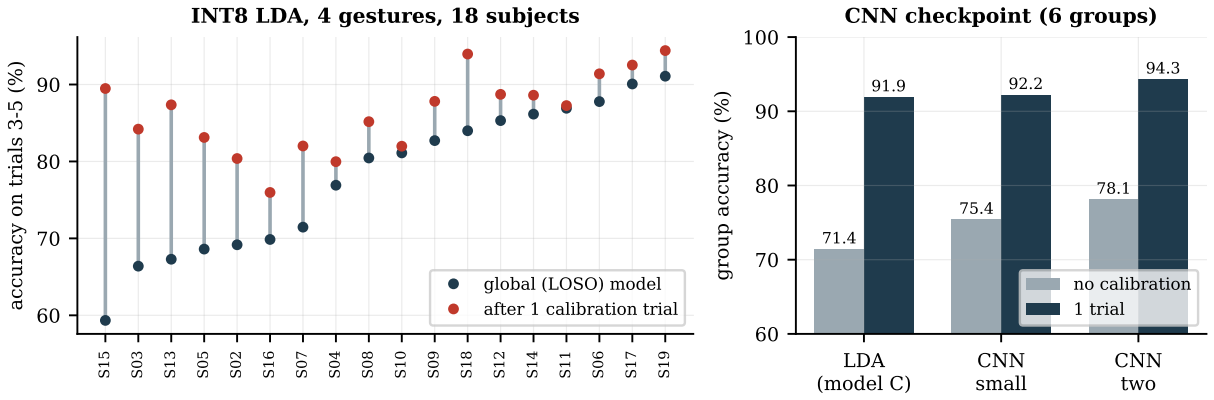


Figure 16: Left: LDA accuracy per subject before and after one calibration trial (four gestures, 18 subjects). Right: the CNN checkpoint, before and after one calibration trial.

gestures and then blending with the global statistics, with the blend chosen on validation. Five ways of calibrating a CNN were compared; only retraining all weights (learning rate 0.001, 20 epochs, chosen on validation) beat the LDA.

After one trial the larger CNN leads by +2.39 pp ([+1.87, +2.90]), better for all 19 subjects; the smaller CNN is level with the LDA (+0.32, [−0.13, +0.76]). Calibration closes most of the CNN's head start (+6.7 pp before, +2.4 pp after), because what the CNN does better, coping with differences between people, is exactly what calibration corrects. The CNN's advantages are real: with one trial it beats the LDA with three, and when calibrated in only one arm position it loses 9.6 pp against the LDA's 12.1 pp (both models still need calibration across positions, as expected from the limb-position effect [6]). But the adoption bar fixed before the study was +3 pp, and the larger CNN needs about 870 times the LDA's arithmetic, a clock of at least 9 MHz, and retraining of about 12,000 weights at every calibration instead of a closed-form blend. **We keep the LDA.**

9 Hardware

9.1 From one chip to a chain of boards

Our first chip served 8 to 24 electrodes from one compute core and up to three 8-channel ADC pods, with a 1 MHz core clock, an 8 MHz SPI clock and a synchroniser between them. Its memories had to be sized for 24 channels (13.5 kbit) even when only 8 were used, and its Stage 2 and 3 took up to 850 of the 1015 cycles of a sample period. The final design instead makes every board identical: one 8-channel pod and one chip. Each board computes features, normalisation and its share of the LDA on its own channels; boards are chained, and only three partial sums per decision cross between them (Figure 17). Compute grows with the number of boards, so one slow clock suffices, and a board needs only 3.7 kbit of memory.

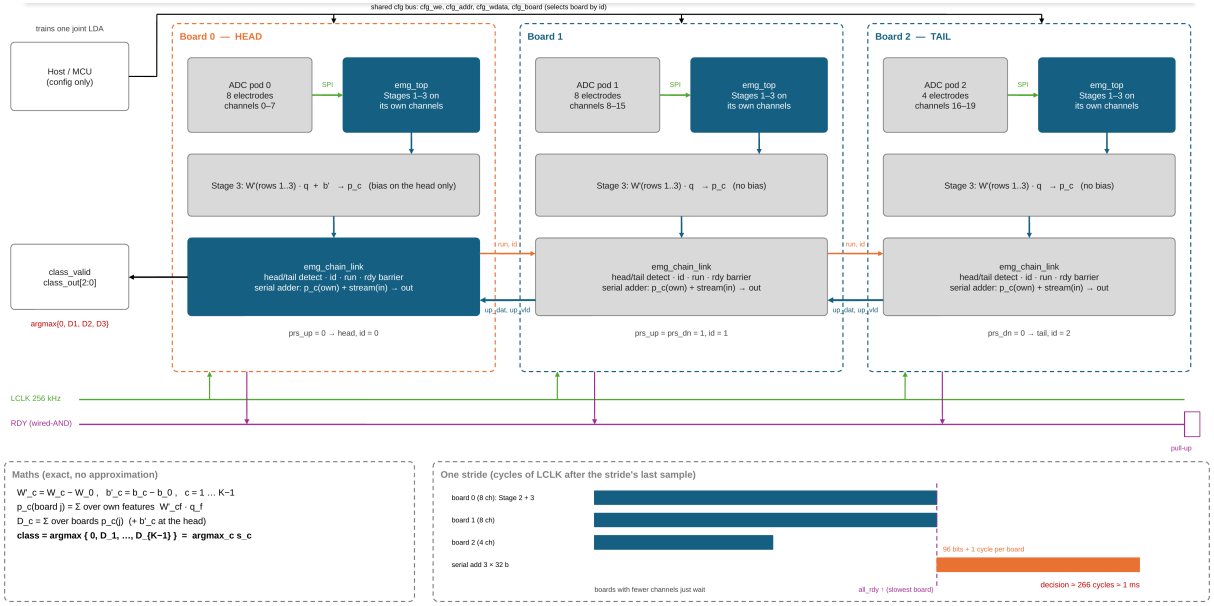


Figure 17: The multi-board system: three boards in a chain (8 + 8 + 4 electrodes), a shared configuration bus, one 256 kHz clock and a wired-AND ready line. Only the partial sums p_c travel along the chain, towards the head.

9.2 One board

Figure 18 shows a board's datapath with the RTL module names.

1. **ADC interface** (`emg_spi_adc`). A timer issues CONVST every 260 cycles; after the pod's data-ready signal, an SPI burst reads 64 bits (eight channels of 8 bits) at half the clock rate.
2. **Stage 1, features** (`emg_feat_accum`, `emg_feat_alu`). For each new frame, a two-stage pipeline reads one channel's history (the previous sample and the sign of the previous difference, 10 bits) and its bank word, computes the five feature increments combinationally and writes the sums back: one channel per cycle, $C \leq 8$ cycles per sample.
3. **Windows.** Two ping-pong banks each accumulate 98 samples. A window is the sum of the two banks, so each completed bank finishes a 196-sample window with a 98-sample stride, using one write pointer and no sliding subtraction (Figure 19). The first decision after reset is masked, because only one bank is full.
4. **Stage 2, normaliser** (`emg_shift_norm`). After every 98th sample it adds the two bank fields, subtracts μ_f, rounds, shifts by k_f and clips, for $f = 0, \ldots, D-1$: $D+2$ cycles.
5. **Stage 3, LDA** (`emg_lda_mac`). One signed 8×8 multiplier and an INT32 accumulator. For each of the $K - 1 = 3$ rows it loads the bias and performs D MACs: $3(D+1)+2$ cycles.
6. **Chain link** (`emg_chain_link`). Position and identity, the common start, the barrier, the bit-serial adder and, on the head, the running argmax (Section 9.5).

All constants (μ_f, k_f, the weight rows and biases) live in writable RAM, because every electrode layout and every calibrated user needs its own model. Every memory has a one-cycle synchronous read, so the RTL maps to SRAM macros or flip-flops unchanged. A board holds 3,656 bits: banks 1,024, μ, k 1,080, weights 960, INT8 features 320, biases 192 and history 80.

9.3 One clock: 256 kHz

The binding constraint is the ADC read. Eight channels of 8 bits at SCLK $= f_{\text{clk}}/2$ take 128 clock cycles per sample, and the sample period is $1/985\,\text{Hz} \approx 1.015\,\text{ms}$. At 128 kHz the read alone would

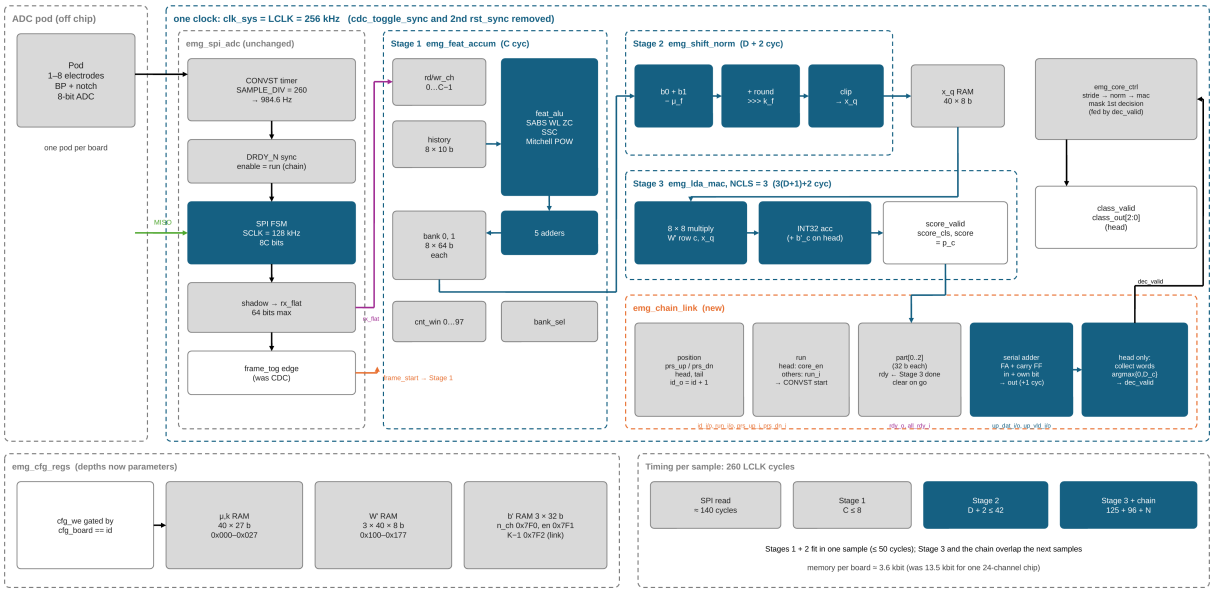


Figure 18: Datapath of one board with the RTL module names.

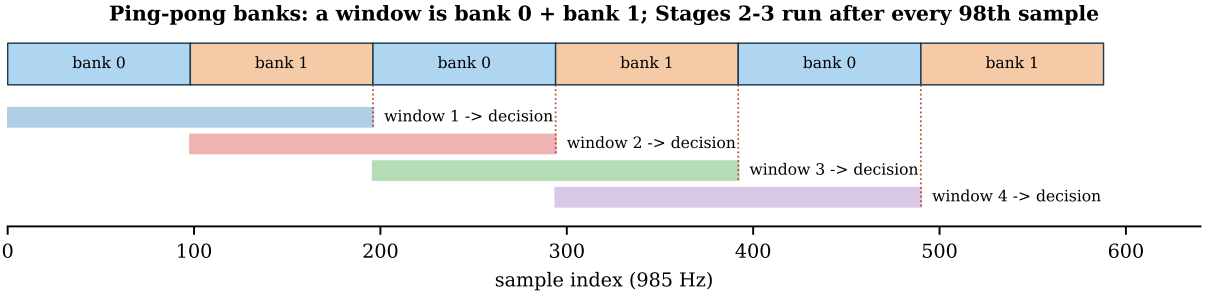


Figure 19: Windowing with two banks. Each bank accumulates 98 samples; a window is the sum of the last two banks, so each completed bank finishes a window and triggers one decision.

fill the whole period, leaving no time for conversion and chip-select set-up. At 256 kHz it takes about 0.5 ms. The sample timer divides by 260, giving 256 kHz/260 = 984.6 Hz, within 0.04% of the dataset's rate. Within one period, Stage 1 needs at most 8 cycles and, after the 98th sample, Stage 2 at most 42, so both finish long before the next frame (Figure 20). Stage 3 and the chain transfer use only the INT8 feature memory and the weights, so they overlap the next samples' reads. With eight channels per board a decision is ready about 280 cycles (≈ 1.1 ms) after the last sample of a stride, out of 25,480 cycles per stride: the datapath is idle about 99% of the time, and one multiplier is enough.

One clock for every block and every board removes all clock-domain crossings, and boards cannot drift apart. Against the single chip's 1 MHz core and 8 MHz interface, the dynamic power of the clock tree and the logic, which scales with frequency, drops accordingly.

9.4 Splitting the LDA across boards

Let board j own the feature set F_j, with $\bigcup_j F_j$ all D features. The single-chip score of class c is a sum over features and therefore splits over boards:

$$s_c = \sum_f W_{cf}\, q_f + b_c = \sum_j \sum_{f \in F_j} W_{cf}\, q_f + b_c. \tag{8}$$

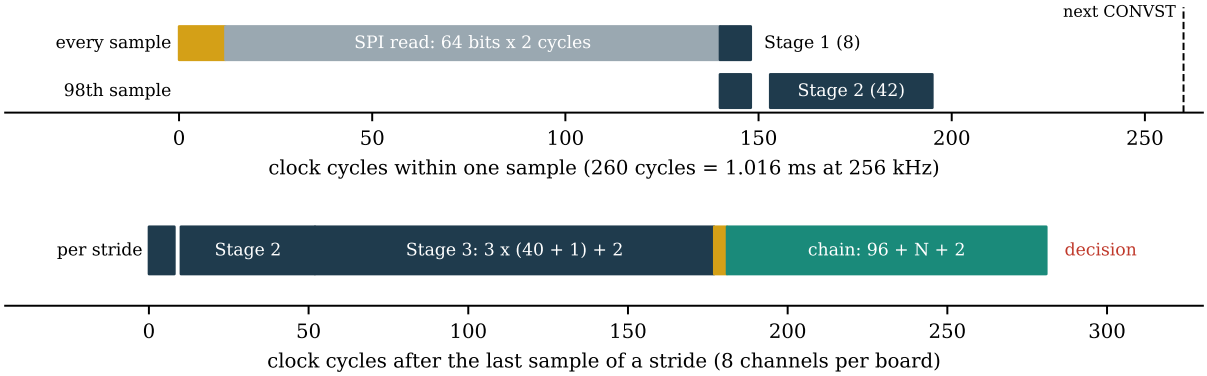


Figure 20: Cycle budget at 256 kHz. Top: one sample period. Bottom: the work after the last sample of a stride.

Only the argmax is needed, and it is unchanged by subtracting the same number from every score:

$$\arg\max_c s_c = \arg\max\{0,\ s_1 - s_0,\ s_2 - s_0,\ s_3 - s_0\}. \tag{9}$$

So each board stores the $K-1$ difference rows $W'_c = W_c - W_0$ $(c = 1, 2, 3)$ for its own features and computes

$$p_c^{(j)} = \sum_{f \in F_j} W'_{cf}\, q_f \quad \big(+\ b'_c = b_c - b_0 \text{ on the head only}\big), \tag{10}$$

$$D_c = \sum_j p_c^{(j)}, \qquad \hat{c} = \arg\max\{0, D_1, D_2, D_3\}. \tag{11}$$

Proposition 1. *If $D_c = s_c - s_0$ exactly for $c = 1, \ldots, K-1$, the chained decision equals the single-chip decision, including ties, when both take the first maximum (strict $>$ in a running comparison).*

Proof. The head starts the running comparison with best value 0 and class 0, which is the value $s_0 - s_0$ of class 0, and then compares $D_1, \ldots, D_{K-1}$ in order with strict $>$. This is the same sequence of comparisons as a running argmax over $s_0, \ldots, s_{K-1}$ shifted by the constant $-s_0$, so it selects the same index. □

In the hardware the difference rows are quantised directly to INT8 with one scale, rather than formed from the INT8 single-chip rows, so the two quantised models are not bit-identical in general; on the real-data test stream of Section 10 they agree on all 79 decisions. The scheme saves a quarter of the work: 120 instead of 160 MACs and weights for four classes, and 96 instead of 128 bits per decision on the chain. Whatever the channel count of a board, it sends 96 bits per decision, compared with 320 bits for its INT8 features or 6,272 bits for its raw samples.

9.5 The chain protocol

The protocol needs no master and no per-board straps; every board runs the same logic.

1. **Position.** Two presence pins tell a board whether a neighbour exists upstream and downstream. A board with nothing upstream is the head, one with nothing downstream the tail. The head has id 0; each board passes id + 1 downstream.
2. **Configuration.** A shared write bus carries address, data and a board number; a board accepts a write only if the number equals its id. A four-plus-four split needs 171 writes (about 1.34 ms).
3. **Start.** The last write sets the head's enable. It travels down the chain as a plain wire (`run`), so every board's sample timer starts in the same clock cycle; with one shared clock the boards then sample the same instant forever.

4. **Barrier.** Each board raises an open-drain ready line when its Stage 3 is done; the line is a wired AND, high only when every board is ready. Its rising edge is `go` on every board in the same cycle, and `ready` clears on `go`. Boards with fewer channels finish earlier and simply wait.
5. **Bit-serial sum.** The tail streams its three partial sums, 32 bits each, least significant bit first. Every board adds its own bit with one full adder and a carry flip-flop and forwards the sum one cycle later; the carry is cleared between words. The head shifts each completed total into a register and compares it with the running best (strict $>$), so the decision is ready $96 + N + 2$ cycles after `go` for N boards.

Least-significant-bit-first order lets the carry ripple naturally, and two's-complement addition modulo 2^{32} is exact as long as every total stays within $\pm 2^{31}$; measured scores are below 10^4. Bit-serial transfer needs two wires per link and a 1-bit adder per board, and takes 0.4 ms of the 99.5 ms stride.

9.6 The design in operation

Figures 21–23 show the simulated waveforms of two boards (four channels each) on a real FORS-EMG stream. At start-up (Figure 21) the reset is released at 20 µs, the boards find their roles and ids from the presence pins, 171 configuration writes load both models, and the head's enable starts both sample timers in the same cycle at 1.38 ms. In steady state (Figure 22) the boards only sample for 98 frames and then run one short burst per stride. Inside a stride (Figure 23) Stage 2 takes 22 cycles, Stage 3 65 cycles, the barrier releases both boards together, 96 bits are added along the chain, and the head decides about 0.75 ms after the last frame. In the stride shown, the totals are $D = (742, -255, 765)$, so the decision is class 3, as in the golden model.

10 Verification

Golden model. A Python model reproduces every integer of Stages 1–3 in two forms, a sample-by-sample model that mirrors the RTL's counters and a vectorised model used for training; the two are cross-checked on every run. Training uses the model's own bit-exact features, so training and hardware see identical numbers.

Chain link alone. A self-checking testbench drives one to four chain links with random partial sums (large, small, strictly negative and forced ties) for 3000 strides each, with $K - 1 = 3$, 2 and 1 rows, and with a random finishing time per board. It computes the expected totals and argmax itself, independently of the Python model, and checks that exactly one decision appears at the head, with the right class, and none elsewhere.

Mutation test. A testbench is trustworthy only if it fails on wrong hardware. Eight bugs were planted one at a time: carry not cleared between words, adder ignoring the carry, ties going to the higher class, last word dropped, decision using a stale best class, class-0 baseline not reset, ready never cleared, ids not incremented. The unit test fails on every one.

Whole chip on real data. One to three boards, each with a behavioural ADC pod, are fed subject 6, trial 1 of FORS-EMG (the most active 1960 samples of each of the four gestures, 7840 samples), with a model trained on the other subjects. For the splits 8, 4+4, 2+6, 6+2, 5+3 and 3+3+2 channels, all 79 decisions equal the golden model bit for bit, and the golden multi-board decisions equal the single-chip LDA on all 79. On windows that contain a single gesture, 72 of 76 decisions are correct; the four errors are the model's own. All 18 checks pass (4 unit, 8 mutants, 6 splits). The single-chip predecessor was verified bit-exactly at 8, 12, 16, 20 and 24 channels with a chip-level cocotb test, and its feature ALU exhaustively on all 196,608 input combinations.

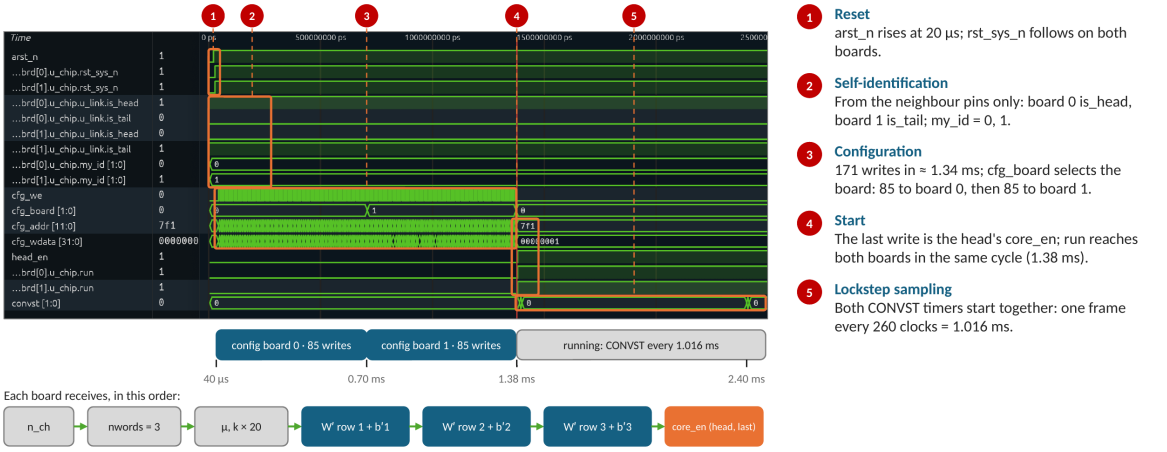


Figure 21: Start-up: reset, self-identification, configuration of board 0 then board 1, and the common start.

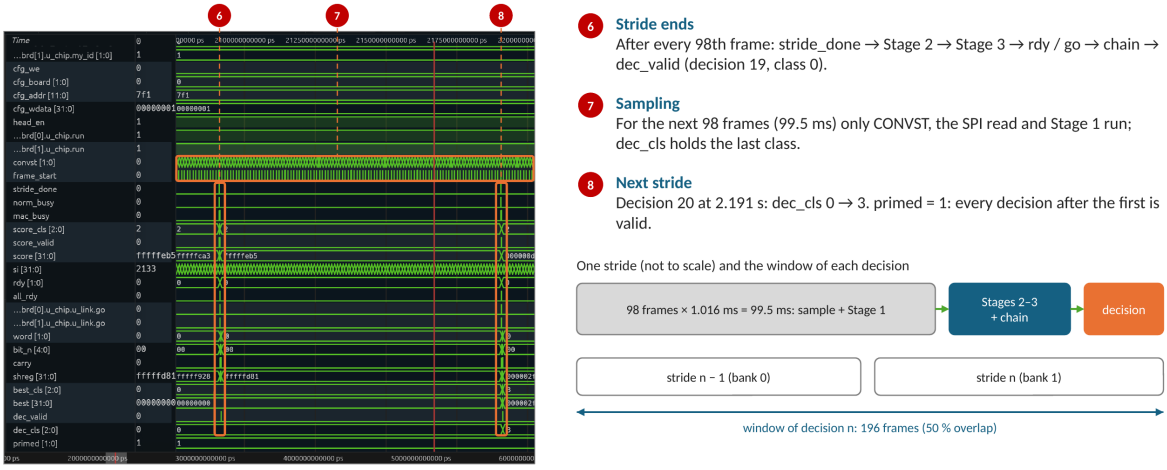


Figure 22: Steady state: continuous sampling and one burst of work per stride (two strides shown).

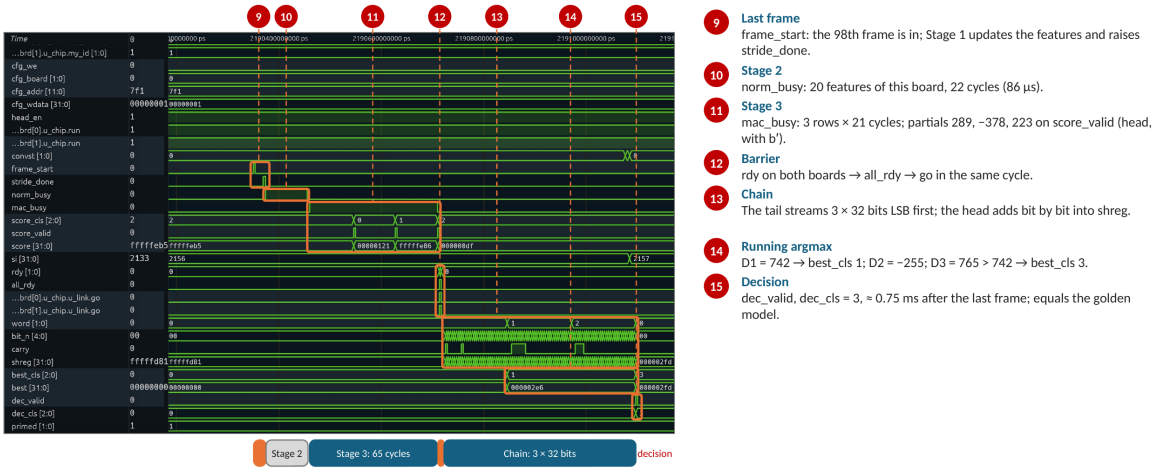


Figure 23: Inside one stride, from the last frame to the decision.

Bugs found. The whole-chip test found one real bug: clearing `ready` on a board's own end of Stage 3 let a fast board raise it again while a slow board's old `ready` was still high, so uneven splits produced two decisions per stride. Clearing on `go` fixed it. A testbench bug (counting the reset transition as a sample) was also found and fixed.

11 Limitations

- **Closed vocabulary.** The classifier always answers one of four gestures and has no rest or reject class; a movement outside the set is forced into one of them. FORS-EMG has no rest recordings.
- **Selection on test folds.** The ranking of gesture subsets used confusion matrices that include the test subjects (Section 4.2), which is mildly optimistic for set D; the merge comparison is free of this.
- **One session per subject.** Day-to-day changes, electrode shift, fatigue and force variation are not measured. Arm-position robustness was measured only in the CNN study.
- **Different settings.** The LDA calibration of Section 8 (four gestures, 18 subjects, fixed $\lambda = 0.5$) and the CNN checkpoint (six groups, held windows, blend chosen on validation, 19 subjects) use different settings; each comparison is paired within its own setting, but their absolute numbers should not be compared with each other.
- **Multiple comparisons.** Paired tests are reported per question without correction for the number of comparisons; every decision rests on intervals well away from zero or on non-inferiority.
- **Simulation only.** The design is verified in RTL simulation with behavioural ADC pods. Area and power in a target process, and tests with real pods and electrodes, remain to be done.

12 Conclusion

For subject-independent sEMG gesture recognition with eight surface electrodes, the vocabulary decides more than the model: choosing gestures driven by distinct, superficial muscles gains 14 pp, and keeping every gesture by merging loses 17 pp. Model capacity adds little: linear models tie with an MLP, and a raw-signal CNN gains at most 2.7 pp before and 2.4 pp after calibration, at three orders of magnitude more arithmetic. An integer-only front end and INT8 weights cost nothing, and one calibration recording per gesture is worth more than any model change. These findings allow a very small machine: identical boards with eight channels, one 256 kHz clock, one

multiplier and 3.7 kbit of memory each, exchanging 96 bits per decision, verified bit-exactly on real data. Next steps are on-chip calibration of the class means, robustness to forearm rotation, a rest class, and synthesis.

A Reproducing the results

All scripts are in the project repository and print their results; they write no files.

Result	Script	Subjects
Gesture sets, classifiers, INT8 loss	`analysis/classifier_classification.py`, `K_4_model_MLP.py`, `K_6_model_LR_comparison.py`	19
Merging versus dropping	`analysis/K_4_LDA_Merge_vs_Drop.py`	19
Feature-type selection	`analysis/fors_emg_feature_selection_lda.py`	18
Integer features H1–H6	`analysis/K_4_LDA_HW_Experiments.py`	18
Chip model vs software LDA	`golden_model/K_4_LDA_Golden_LOSO.py`	18
CNNs vs LDA	`analysis/K_4_CNN_LOSO.py`	19
LDA and CNN calibration	`analysis/K_4_LDA_Calib.py`, `K_4_CNN_Calib.py`	18
Multi-board RTL verification	`RTL/tb/verify_all.py`	real data, subject 6

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